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Exhaust-gas treatment arrangement

Abstract

An exhaust-gas treatment arrangement for an exhaust system of an internal-combustion engine includes a diesel internal-combustion engine. The arrangement includes a first exhaust-gas treatment unit and, downstream of the first exhaust-gas treatment unit, an exhaust-gas treatment assembly. The first exhaust-gas treatment unit and the exhaust-gas treatment assembly are arranged axially succeeding one another in the direction of a longitudinal axis of a flow path encompassing the first exhaust-gas treatment unit and the exhaust-gas treatment assembly. A hydrocarbon-feeding assembly is provided for feeding hydrocarbon into exhaust gas flowing in the flow path. The hydrocarbon-feeding assembly includes a hydrocarbon-dispensing unit dispensing hydrocarbon into the flow path downstream of the first exhaust-gas treatment unit and upstream of the exhaust-gas treatment assembly. A swirl-flow-generating unit is provided in the flow path upstream of the hydrocarbon-dispensing unit.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority of German patent application no. 10 2022 132 127.4, filed Dec. 5, 2022, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to an exhaust-gas treatment arrangement for an exhaust system of an internal-combustion engine, in particular a diesel internal-combustion engine.

BACKGROUND

(3) For the purpose of attaining the permissible exhaust-gas values of the exhaust gases emitted by internal-combustion engines, in particular by diesel internal-combustion engines, it is known, for instance, to inject a reagent such as, for example, a urea/water solution into the exhaust gas, in order to lower the content of nitrogen oxide in an exhaust-gas treatment unit taking the form, in particular, of an SCR catalytic-converter unit. It is also known to combine various types of catalytic converter, such as, for example, SCR catalytic-converter units and oxidation catalytic-converter units, in an exhaust-gas treatment arrangement of such a type, in order to obtain a diminution that is as efficient as possible of the pollutant content in the exhaust gas emitted to the environment.

(4) For the catalytic reactions to be carried out in various system regions of an exhaust-gas treatment arrangement of such a type, it is necessary that the catalyst materials that are active for this purpose have a sufficiently high temperature. Since the heat transported in the exhaust gas is often not sufficient, particularly in start-up phases of the working operation of an internal-combustion engine or at comparatively low ambient temperatures, to attain quickly or to maintain reliably a sufficiently high temperature for the catalytic reactions to be carried out, it is, for instance, known to integrate electrically operated exhaust-gas heaters into an exhaust system, in order to transfer heat to the exhaust gas emitted by an internal-combustion engine, or to a different gas fed into the exhaust system, upstream of one or more catalytically active system regions, which is then transported by the exhaust gas, or by the gas, to the catalytically active system region(s) and transferred thereto.

SUMMARY

(5) It is an object of the present disclosure to provide an exhaust-gas treatment arrangement for an exhaust system of an internal-combustion engine, in particular a diesel internal-combustion engine, with which, with a structurally simple and compact configuration, a reliable heating of system regions provided for the purpose of exhaust-gas treatment can be guaranteed.

(6) In accordance with the disclosure, this object is, for example, achieved via an exhaust-gas treatment arrangement for an exhaust system of an internal-combustion engine, in particular a diesel internal-combustion engine, including at least one first exhaust-gas treatment unit and, downstream of the at least one first exhaust-gas treatment unit, an exhaust-gas treatment assembly, wherein the at least one first exhaust-gas treatment unit and the exhaust-gas treatment assembly are arranged axially succeeding one another in the direction of a longitudinal axis of a flow path encompassing the at least one first exhaust-gas treatment unit and the exhaust-gas treatment assembly, wherein a hydrocarbon-feeding assembly has been provided for the purpose of feeding hydrocarbon into exhaust gas flowing in the flow path, wherein the hydrocarbon-feeding assembly includes a hydrocarbon-dispensing unit, for dispensing hydrocarbon into the flow path downstream of the at least one first exhaust-gas treatment unit and upstream of the exhaust-gas treatment assembly, and a swirl-flow-generating unit in the flow path upstream of the hydrocarbon-dispensing unit.

(7) By virtue of the provision of the hydrocarbon-feeding assembly constructed in accordance with the disclosure, the possibility is created of feeding hydrocarbon releasing heat in the course of its

oxidation and consequently contributing to the heating of system regions provided for the purpose of exhaust-gas treatment—that is, for instance, the fuel that is also to be supplied to an internal-combustion engine—into the stream of exhaust gas, and of efficiently intermixing it with exhaust gas. In particular, the swirl-flow-generating unit positioned in the flow path upstream of the location of the feeding of the hydrocarbon into the flow path, which transmits a peripheral-flow-direction component to the stream of exhaust gas flowing in the flow path, contributes to the efficient intermixing of exhaust gas and hydrocarbon, so that a flow-direction component in the peripheral direction is superimposed on a main flow direction of exhaust gas in the flow path which is oriented substantially in the direction of the longitudinal axis of the flow path. This results in occurrences of swirling, and basically in a longer flow path between the at least one first exhaust-gas treatment unit and the exhaust-gas treatment assembly, which is available and is utilized for the purpose of intermixing exhaust gas and hydrocarbon.

(8) The hydrocarbon-dispensing unit may be supported on a flow-path housing axially between an outlet region of the at least one first exhaust-gas treatment unit and an inlet region of the exhaust-gas treatment assembly, so that the hydrocarbon-dispensing unit can be accommodated in the exhaust-gas treatment arrangement without basically protruding beyond the outer contour thereof.

(9) For the purpose of further assisting the efficient intermixing of exhaust gas and hydrocarbon, the hydrocarbon-dispensing unit for dispensing hydrocarbon may have been arranged with a main dispensing direction substantially orthogonal to the longitudinal axis of the flow path or oriented substantially tangentially with respect to the longitudinal axis of the flow path.

(10) For the purpose of generating the swirl current, the swirl-flow-generating unit may include a plurality of current-deflecting elements.

(11) For an efficient and compact configuration of the swirl-flow-generating unit, the current-deflecting elements may have been arranged succeeding one another in the peripheral direction about the longitudinal axis of the flow path, or/and they may have been arranged extending substantially radially with respect to the longitudinal axis of the flow path, or/and they may have been arranged in a star-like configuration.

(12) For a stable configuration, there may be provision that the swirl-flow-generating unit includes a ring-like body supported on the flow path, and the current-deflecting elements adjoin the ring-like body in their radial outer end region, or/and that the current-deflecting elements adjoin one another in their radial inner end region, or/and that the swirl-flow-generating unit takes the form of a sheet-metal formed part.

(13) For a compact configuration of the exhaust-gas treatment arrangement, particularly in the axial direction, the swirl-flow-generating unit may have been arranged on an outlet region of the at least one first exhaust-gas treatment unit.

(14) For the purpose of further lessening the pollutant component contained in the exhaust gas emitted by an internal-combustion engine, at least one second exhaust-gas treatment unit may have been arranged downstream of the exhaust-gas treatment assembly.

(15) In order to be able to carry out a catalytic cleanup reaction efficiently in the at least one second exhaust-gas treatment unit, an exhaust-gas/reagent mixing section with a mixing duct elongated substantially in the direction of a longitudinal axis of the mixing section and with a reagent-dispensing unit for dispensing reagent into the mixing duct may have been provided downstream of the exhaust-gas treatment assembly and upstream of the at least one second exhaust-gas treatment unit.

(16) For a compact configuration of the exhaust-gas treatment arrangement, it is proposed that the longitudinal axis of the mixing section is substantially parallel to the longitudinal axis of the flow path, or/and that the flow path and the exhaust-gas/reagent mixing section overlap one another substantially completely in the axial direction, so that an inlet region of the exhaust-gas/reagent mixing section in the direction of the longitudinal axis of the mixing section is positioned substantially in the same axial region as an outlet region of the exhaust-gas treatment assembly, and

an outlet region of the exhaust-gas/reagent mixing section in the direction of the longitudinal axis of the mixing section is positioned substantially in the same axial region as an outlet region of the at least one first exhaust-gas treatment unit.

(17) An efficient utilization of the construction volume provided for the exhaust-gas treatment arrangement can be obtained if a main flow direction of exhaust gas in the mixing duct is directed substantially contrary to a main flow direction of exhaust gas in the flow path.

(18) Furthermore, for this purpose the at least one second exhaust-gas treatment unit may be elongated in the direction of a longitudinal axis of the exhaust-gas treatment unit, which is substantially parallel to the longitudinal axis of the flow path, and may be capable of being flowed through substantially in the direction of the longitudinal axis of the exhaust-gas treatment unit, and the at least one second exhaust-gas treatment unit and the exhaust-gas/reagent mixing section may overlap one another substantially completely in the axial direction, so that the inlet region of the exhaust-gas/reagent mixing section in the direction of the longitudinal axis of the mixing section is positioned substantially in the same axial region as an outlet region of the at least one second exhaust-gas treatment unit, and the outlet region of the exhaust-gas/reagent mixing section in the direction of the longitudinal axis of the mixing section is positioned substantially in the same axial region as an inlet region of the at least one second exhaust-gas treatment unit.

(19) For the purpose of providing a flow connection, the inlet region of the exhaust-gas/reagent mixing section may have been connected to the outlet region of the exhaust-gas treatment assembly via a first current-deflecting housing, and the outlet region of the exhaust-gas/reagent mixing section may have been connected to the inlet region of the at least one second exhaust-gas treatment unit via a second current-deflecting housing.

(20) For an increase in the efficiency of exhaust-gas treatment with a compact style of construction, at least two second exhaust-gas treatment units that are capable of being flowed through parallel to one another may have been provided at right angles to the longitudinal axis of the exhaust-gas treatment unit, alongside one another and overlapping one another substantially completely in the direction of the longitudinal axis of the exhaust-gas treatment unit.

(21) At least one exhaust-gas treatment unit may include at least one SCR catalytic-converter unit or/and at least one ammonia-blocking catalytic-converter unit. Furthermore, the exhaust-gas treatment assembly may include an oxidation catalytic-converter unit or/and a particle-filter unit.

(22) For the most uniform possible feeding of exhaust gas intermixed with hydrocarbon over the cross-section of the exhaust-gas treatment assembly, a flow-distribution element may have been arranged in the flow path downstream of the hydrocarbon-dispensing unit and upstream of the exhaust-gas treatment assembly.

(23) The flow-distribution element may, for instance, take the form of a sheet-metal formed part in the manner of a plate, and may cover an inlet region of the exhaust-gas treatment assembly substantially completely. A large number of flow-through openings may have been provided in the flow-distribution element. These flow-through openings may have been formed with substantially the same cross-section for a uniform feeding of exhaust gas or hydrocarbon into the exhaust-gas treatment assembly, and may have been distributed substantially uniformly over the flow-distribution element.

(24) The disclosure further relates to an exhaust system for an internal-combustion engine, in particular a diesel internal-combustion engine, including at least one exhaust-gas treatment arrangement constructed in accordance with the disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The invention will now be described with reference to the drawings wherein:

- (2) FIG. 1 shows a side view, represented partially open, of an exhaust-gas treatment unit for an internal-combustion engine;
- (3) FIG. 2 shows, in schematic and developed representation, the system regions of the exhaust-gas treatment arrangement shown in FIG. 1, succeeding one another in the direction of flow and overlapping one another substantially axially;
- (4) FIG. 3 shows a schematic cross-sectional view of the exhaust-gas treatment arrangement shown in FIG. 1, cut along a line III-III in FIG. 1; and,
- (5) FIG. 4 shows a perspective cross-sectional view of the exhaust-gas treatment arrangement shown in FIG. 1, cut along a line IV-IV in FIG. 1.

DETAILED DESCRIPTION

- (6) In FIG. 1 an exhaust-gas treatment arrangement **10** for an exhaust system **12**, in particular for a diesel internal-combustion engine, can be discerned in side view. An exhaust-gas treatment arrangement **10** of such a type may, for instance, have been provided in an exhaust system **12** of a utility motor vehicle or of a truck. In FIGS. 2 and 3 the exhaust-gas treatment arrangement **10** is represented in schematic view, FIG. 2 representing a peripheral embodiment of the exhaust-gas treatment arrangement **10**, basically elongated in the direction of a longitudinal axis L.sub.1 of the exhaust-gas treatment arrangement.
- (7) The exhaust-gas treatment arrangement **10** includes a flow path, denoted generally by **14** and elongated in the direction of a longitudinal axis L.sub.2 of the flow path, with a substantially tubular flow-path housing **16** which is in one piece or composed of several parts. At an upstream end region **18** of the flow path **14** a feed housing **20** adjoins the flow-path housing **16**. Exhaust gas A emitted by an internal-combustion engine is fed into the exhaust-gas treatment arrangement **10**, or into the flow path **14**, via the feed housing **20**. In a downstream end region **22** of the flow path **14** a first deflecting housing **24** adjoins the flow-path housing **16**. In the first deflecting housing **24** the exhaust gas A flowing through, or emerging from, the flow path **14** is deflected by approximately 180° and fed into an exhaust-gas/reagent mixing section **26**. The exhaust-gas/reagent mixing section **26** includes, in a tubular mixing-section housing **28**, a mixing duct **30** elongated in the direction of a longitudinal axis L.sub.3 of the mixing section. At an upstream end region **32** of the exhaust-gas/reagent mixing section **26** a reagent-dispensing unit **34**, also generally designated as an injector, has been provided, supported, for instance, on the first deflecting housing **24**, through which a reagent R, for instance a urea/water solution, is injected into the mixing duct **30**. For the purpose of assisting the intermixing of exhaust gas A and reagent R, a mixer **36**, including, for instance, a plurality of deflecting blades or such like, may have been arranged in the mixing-section housing **28**.
- (8) In a downstream end region **37** of the exhaust-gas/reagent mixing section **26** the mixing-section housing **28** adjoins a second deflecting housing **38**. In the second deflecting housing **38** a deflection of the stream of exhaust gas by approximately 180° again takes place. In the embodiment represented, the exhaust gas is fed via the second deflecting housing **38** into two second exhaust-gas treatment units **40**, **42** which are capable of being flowed through parallel to one another. Each of the second exhaust-gas treatment units **40**, **42** includes a tubular exhaust-gas treatment-unit housing **44**, **46** elongated in the direction of a respective longitudinal axis L.sub.4, L.sub.5 of the exhaust-gas treatment unit. A respective upstream end region of the exhaust-gas treatment-unit housings **44**, **46** provides a respective inlet region **48**, **50** of the second exhaust-gas treatment units **40**, **42**, and a respective downstream end region of the exhaust-gas treatment-unit housings **44**, **46** provides a respective outlet region **52**, **54** of the second exhaust-gas treatment units **40**, **42**. These are open to a drainage housing **56** via which the exhaust gas A treated in the exhaust-gas treatment arrangement **10** leaves the exhaust-gas treatment arrangement **10** to reach further system regions of the exhaust system **12**, for instance one or more mufflers or such like.
- (9) FIGS. 1 and 3 show that in the exhaust-gas treatment arrangement **10** the flow path **14**, the exhaust-gas/reagent mixing section **26** and the second exhaust-gas treatment units **40**, **42** are

positioned in such a manner that longitudinal axes L.sub.2, L.sub.3, L.sub.4, L.sub.5 thereof are substantially parallel to one another and to the longitudinal axis L.sub.1 of the exhaust-gas treatment arrangement, and that they overlap one another substantially completely in the axial direction. This means that an inlet region **58** of the exhaust-gas/reagent mixing section **26** is situated substantially in the same axial region as an outlet region **60** of an exhaust-gas treatment assembly **62** arranged in the flow path **14**, and that an outlet region **64** of the exhaust-gas/reagent mixing section **26** is situated substantially in the same axial region as an inlet region **66** of a first exhaust-gas treatment unit **68** arranged in the flow path **14**. Equally, the outlet region **64** of the exhaust-gas/reagent mixing section **26** is situated substantially in the same axial region as the inlet regions **48**, **50** of the second exhaust-gas treatment units **40**, **42**, whereas the inlet region **58** of the exhaust-gas/reagent mixing section **26** is situated substantially in the same axial region as the outlet regions **52**, **54** of the second exhaust-gas treatment units **40**, **42**.

(10) In the embodiment represented, the first exhaust-gas treatment unit **68** arranged in the flow path **14**, or in the flow-path housing **16**, includes an SCR catalytic-converter unit **70** and an ammonia-blocking catalytic-converter unit **72**, succeeding one another in the direction of flow or axially in the direction of the longitudinal axis L.sub.2 of the flow path. The exhaust gas A, fed into the flow path **14** via the feed housing **20**, flows through the two catalytic-converter units **70**, **72** of the first exhaust-gas treatment unit **68** substantially in a main flow direction H.sub.1 of exhaust gas in the flow path **14**, the main flow direction H.sub.1 of exhaust gas in the flow path **14** being oriented substantially parallel to the longitudinal axis L.sub.2 of the flow path. Of course, flow-direction components deviating from this main flow direction H.sub.1 of exhaust gas may be present, for instance in regions in which occurrences of swirling or turbulence arise. In the exhaust system **12** a further reagent-dispensing unit may have been arranged upstream of the exhaust-gas treatment arrangement **10** or of the SCR catalytic-converter unit **70**, in order to feed a reagent, for instance a urea/water solution, into the stream of exhaust gas upstream of the SCR catalytic-converter unit **70**.

(11) Downstream of the two catalytic-converter units **70**, **72** of the first exhaust-gas treatment unit **68** an oxidation catalytic-converter unit **74**, in particular a diesel oxidation catalytic-converter unit, and a particle-filter unit **76** of the exhaust-gas treatment assembly **62** are arranged, succeeding one another in the direction of flow. The exhaust gas A flowing through the flow path **14** flows substantially in the main flow direction H.sub.1 of exhaust gas, firstly through the oxidation catalytic-converter unit **74** and then through the particle-filter unit **76**, before it is deflected by the first deflecting housing **24** in the direction toward the exhaust-gas/reagent mixing section **26**.

(12) Attention is drawn to the fact that in the case of the exhaust-gas treatment arrangement **10** both the first exhaust-gas treatment unit **68** and the exhaust-gas treatment assembly **62** may have been constructed differently than in the embodiment represented. For instance, the first exhaust-gas treatment unit **68** might include only the SCR catalytic-converter unit **70**. The exhaust-gas treatment assembly **62** might, for instance, include only the oxidation catalytic-converter unit **74**.

(13) The exhaust-gas treatment arrangement **10** includes, assigned to the flow path, a hydrocarbon-feeding assembly denoted generally by **78**. This assembly includes a hydrocarbon-dispensing unit **80**, also generally designated as an injector, which is positioned on the flow-path housing **16** axially between the at least one first exhaust-gas treatment unit **68** and the exhaust-gas treatment assembly **62**.

(14) The hydrogen-dispensing unit **80** inputs hydrocarbon K—for instance, the fuel also used in an internal-combustion engine—in the form of a spray mist or in droplet form into the flow path **14** in a region downstream of the at least one first exhaust-gas treatment unit **68** and upstream of the exhaust-gas treatment assembly **62**. In this process, the hydrocarbon K can be fed into the flow path **14** with a main dispensing direction H.sub.2, discernible in FIG. 2, which is oriented substantially orthogonally or tangentially to the longitudinal axis L.sub.2 of the flow path—that is, it may have a peripheral-orientation component.

(15) The hydrocarbon-feeding assembly **78** further includes a swirl-flow-generating unit **82** downstream of the first exhaust-gas treatment unit **68**, in particular covering an outlet region **84** thereof. This swirl-flow-generating unit **82**, discernible in FIG. **4**, may take the form, for instance, of a sheet-metal formed part and may include a plurality of current-deflecting elements **86** extending substantially radially with respect to the longitudinal axis L.sub.2 of the flow path. The current-deflecting elements **86** may have been formed in the swirl-flow-generating unit **82** as axial moldings, between which openings **96** extending substantially radially, for instance slit-like openings, have been formed. In the region of these openings **96** the exhaust gas flowing through the first exhaust-gas treatment unit **68** is able to flow into the region of the flow path **14** into which the hydrocarbon K is fed. In a radially outer region, the current-deflecting elements **86** adjoin a ring-like body **88** of the swirl-flow-generating unit **82**, with which the latter can be supported, for instance on the inner periphery of the flow-path housing **16** or at the downstream end of the at least one first exhaust-gas treatment unit **68**. The current-deflecting elements **86** adjoin one another radially on the inside, so that a basically very stable structure of the swirl-flow-generating unit **82**, taking the form, for instance, of a sheet-metal formed part, is obtained.

(16) The current-deflecting elements **86** provide deflecting surfaces positioned with respect to the longitudinal axis L.sub.2 of the flow path, which deflect the exhaust gas emerging from the first exhaust-gas treatment unit **68** substantially in the main flow direction H.sub.1 of exhaust gas in the peripheral direction, so that this exhaust gas has a peripheral-flow-direction component superimposed on the main flow direction H.sub.1 of exhaust gas. The hydrocarbon K is injected into this swirl current of the exhaust gas A by the hydrocarbon-dispensing unit **80** and is consequently carried along by the swirl current to the current in the peripheral direction or even in the axial direction. As a result, an efficient intermixing of hydrocarbon K and exhaust gas A is obtained in the flow path in the volume region between the outlet region **84** of the first exhaust-gas treatment unit **68** and an inlet region **90** of the exhaust-gas treatment assembly **62**.

(17) In order to obtain a feeding that is as uniform as possible of the exhaust gas A permeated with hydrocarbon K into the exhaust-gas treatment assembly **62**, in particular into the oxidation catalytic-converter unit **74**, a plate-like flow-distribution element **92** is arranged upstream of the exhaust-gas treatment assembly **62**, covering an inlet region **90** of the same substantially completely. This flow-distribution element **92**, taking the form, for instance, of a sheet-metal formed part, is perforated—that is, it exhibits a large number of flow-through openings **94** through which the mixture including exhaust gas A and hydrocarbon K enters the exhaust-gas treatment assembly **62**, in particular the oxidation catalytic-converter unit **74**. The flow-through openings **94** are distributed substantially uniformly over the surface of the flow-distribution element **92** and have substantially identical opening cross-sections relative to one another.

(18) By oxidation of the hydrocarbon K transported in the exhaust gas A in the oxidation catalytic-converter unit **74**, heat is released which, on the one hand, contributes to a heating of the oxidation catalytic-converter unit **74** and consequently ensures that the latter is quickly brought to an operating temperature required for the catalytic reaction, or is efficiently maintained at this temperature. On the other hand, some of this heat can be carried by the exhaust gas flowing through the oxidation catalytic-converter unit **74** to the following system regions, in particular to the exhaust-gas/reagent mixing section **26** and to the second exhaust-gas treatment units **40**, **42**. In the exhaust-gas/reagent mixing section **26** this can contribute to an intensified evaporation of the reagent R injected in liquid form. In the second exhaust-gas treatment units **40**, **42**, respective SCR catalytic-converter units **98**, **100** of the same can be efficiently heated and brought to the temperature required for the SCR reaction more quickly, or can be maintained at this temperature.

(19) With the structure, according to the disclosure, of an exhaust-gas treatment assembly, it is ensured, on the one hand by the admixture of hydrocarbon into the stream of exhaust gas, and on the other hand by an efficient intermixing of the hydrocarbon with the exhaust gas flowing in the exhaust-gas treatment assembly, that as a result of oxidation of the hydrocarbon, distributed over a

large area, heat that is capable of being utilized for heating various system regions of the exhaust-gas treatment arrangement **10** provided for the purpose of carrying out catalytic reactions can be released substantially uniformly. By virtue of the integration of the hydrocarbon-feeding assembly into the flow path in the manner described in the foregoing, a compact configuration of the exhaust-gas treatment arrangement is obtained or retained, in which the flow principle is guaranteed, namely that the main flow direction H.sub.1 of exhaust gas in the flow path, also encompassing the hydrocarbon-feeding assembly, is directed substantially contrary to a main flow direction H.sub.4 of exhaust gas in the exhaust-gas/reagent mixing section, whereas a main flow direction H.sub.5 of exhaust gas in the second exhaust-gas treatment units is directed to be identical to the main flow direction H.sub.1 of exhaust gas in the flow path but is directed contrary to the main flow direction H.sub.4 of exhaust gas in the exhaust-gas/reagent mixing section. Hence it can be guaranteed that, with an axially very compact style of construction, by reason of the feeding of hydrocarbon and by reason of the efficient intermixing of the hydrocarbon with the exhaust gas the statutory limiting values for pollutant contents in the exhaust gas can be attained or maintained under highly diverse operating circumstances of an internal-combustion engine or of the exhaust system.

(20) It is understood that the foregoing description is that of the preferred embodiments of the invention and that various changes and modifications may be made thereto without departing from the spirit and scope of the invention as defined in the appended claims.

Claims

1. An exhaust-gas treatment arrangement for an exhaust gas system of an internal combustion engine including a diesel internal combustion engine, the exhaust-gas treatment arrangement defining a flow path and comprising: a first exhaust-gas treatment unit; an exhaust-gas treatment assembly downstream of said first exhaust-gas treatment unit; said first exhaust-gas treatment unit and said exhaust-gas treatment assembly being arranged axially one after the other in a direction of a longitudinal axis (L.sub.2) of said flow path with said flow path including said first exhaust-gas treatment unit and said exhaust-gas treatment assembly; a hydrocarbon feeding assembly for introducing hydrocarbon into exhaust gas flowing in said flow path; said hydrocarbon feeding assembly including a hydrocarbon-dispensing unit for dispensing hydrocarbon into said flow path downstream of said first exhaust-gas treatment unit and upstream of said exhaust-gas treatment assembly; a swirl-flow-generating unit arranged in said flow path upstream of said hydrocarbon-dispensing unit; a second exhaust-gas treatment unit arranged downstream of said exhaust-gas treatment assembly; and, an exhaust-gas/reagent mixing section defining a longitudinal axis (L.sub.3) and having a mixing duct elongated substantially in a direction of said longitudinal axis (L.sub.3) of the mixing section; said exhaust-gas/reagent mixing section further having a reagent-dispensing unit for dispensing reagent into said mixing duct and being disposed downstream of said exhaust-gas treatment assembly and upstream of said second exhaust-gas treatment unit.
2. The exhaust-gas treatment arrangement of claim 1, further comprising a flow-path housing accommodating said flow path therein; and, said hydrocarbon-dispensing unit being supported on said flow-path housing axially between an outlet region of said first exhaust-gas treatment unit and an inlet region of said exhaust-gas treatment assembly.
3. The exhaust-gas treatment arrangement of claim 2, wherein said hydrocarbon-dispensing unit is arranged with a main dispensing direction (H.sub.2) which is substantially orthogonal to said longitudinal axis (L.sub.2) of said flow path or oriented substantially tangentially with respect to said longitudinal axis (L.sub.2) of said flow path.
4. The exhaust-gas treatment arrangement of claim 1, wherein said swirl-flow-generating unit includes a plurality of current-deflecting elements.
5. The exhaust-gas treatment arrangement of claim 4, wherein at least one of the following applies:

- i) said current-deflecting elements are arranged succeeding one another in a peripheral direction about said longitudinal axis (L.sub.2) of said flow path; and, ii) said current-deflecting elements are arranged extending substantially radially with respect to said longitudinal axis (L.sub.2) of said flow path; and, iii) said current-deflecting elements are arranged in a star-like configuration.
6. The exhaust-gas treatment arrangement of claim 4, wherein said swirl-flow-generating unit includes a ring-like body supported on said flow path; and, wherein at least one of the following applies: i) said current-deflecting elements adjoin said ring-like body in a radial outer end region thereof; and, ii) said current-deflecting elements adjoin one another in a radial inner end region thereof; and, iii) said swirl-flow-generating unit assumes the form of a sheet-metal formed part.
7. The exhaust-gas treatment arrangement of claim 1, wherein said swirl-flow-generating unit is arranged on an outlet region of said first exhaust-gas treatment unit.
8. The exhaust-gas treatment arrangement of claim 1, wherein at least one of the following applies: i) said longitudinal axis (L.sub.3) of said mixing section is substantially parallel to said longitudinal axis (L.sub.2) of said flow path; and, ii) said flow path and said exhaust-gas/reagent mixing section overlap one another substantially completely in axial direction, so that an inlet region of the exhaust-gas/reagent mixing section in said direction of said longitudinal axis (L.sub.3) of said mixing section is positioned substantially in a same axial region as an outlet region of said exhaust-gas treatment assembly; and, iii) an outlet region of said exhaust-gas/reagent mixing section in said direction of said longitudinal axis (L.sub.3) of said mixing section is positioned substantially in a same axial region as an inlet region of said first exhaust-gas treatment unit.
9. The exhaust-gas treatment arrangement of claim 1, wherein a main flow direction (H.sub.4) of exhaust gas in said mixing duct is directed substantially opposite to a main flow direction (H.sub.1) of exhaust gas in the flow path.
10. The exhaust-gas treatment arrangement of claim 1, wherein said second exhaust-gas treatment unit is elongated in a direction of a longitudinal axis (L.sub.4, L.sub.5) of said exhaust-gas treatment unit which is substantially parallel to said longitudinal axis (L.sub.2) of said flow path and is configured to be flowed through substantially in the direction of the longitudinal axis (L.sub.4, L.sub.5) of said second exhaust-gas treatment unit; and, said second exhaust-gas treatment unit and said exhaust-gas/reagent mixing section overlap one another substantially completely in an axial direction, so that an inlet region of said exhaust-gas/reagent mixing section is positioned in said direction of the longitudinal axis (L.sub.3) of said mixing section substantially in a same axial region as an outlet region of said second exhaust-gas treatment unit, and an outlet region of said exhaust-gas/reagent mixing section is positioned in a direction of said longitudinal axis (L.sub.3) of said mixing section substantially in the same axial region as an inlet region of said second exhaust-gas treatment unit.
11. The exhaust-gas treatment arrangement of claim 10, wherein said inlet region of said exhaust-gas/reagent mixing section is connected to the outlet region of said exhaust-gas treatment assembly via a first current-deflecting housing, and the outlet region of the exhaust-gas/reagent mixing section is connected to the inlet region of said second exhaust-gas treatment unit via a second current-deflecting housing.
12. The exhaust-gas treatment arrangement of claim 10, further comprising an additional second exhaust gas treatment unit to provide two second exhaust-gas treatment units, which are configured to be flowed through parallel to one another, and are provided transverse to the longitudinal axis (L.sub.4, L.sub.5) of the second exhaust-gas treatment units, alongside one another and overlapping one another substantially completely in the direction of the longitudinal axis (L.sub.4, L.sub.5) of the second exhaust-gas treatment units.
13. The exhaust-gas treatment arrangement of claim 1, wherein at least one of said exhaust-gas treatment units includes at least one SCR catalytic-converter unit and/or at least one ammonia-blocking catalytic-converter unit and/or wherein said exhaust-gas treatment assembly includes an oxidation catalytic-converter unit and/or a particle-filter unit.

14. The exhaust-gas treatment arrangement of claim 1, further comprising a flow-distribution element arranged in said flow path downstream of the hydrocarbon-dispensing unit and upstream of said exhaust-gas treatment assembly.

15. The exhaust-gas treatment arrangement of claim 14, wherein said flow-distribution element is formed in the manner of a plate and covers an inlet region of said exhaust-gas treatment assembly substantially completely, wherein a plurality of flow-through openings are formed with substantially the same cross-section and substantially uniformly distributed over the flow-distribution element, are provided in the flow-distribution element.

16. The exhaust-gas treatment arrangement of claim 15, wherein said flow-through openings are formed with substantially the same cross-section and substantially uniformly distributed over the flow-distribution element.

17. An exhaust system for an internal-combustion engine including a diesel internal-combustion engine, the exhaust system comprising: an exhaust-gas conduit conducting exhaust gas away from the combustion engine; and, an exhaust-gas treatment arrangement communicating with said exhaust-gas conduit and defining a flow path and said exhaust-gas treatment arrangement including: a first exhaust-gas treatment unit; an exhaust-gas treatment assembly downstream of said first exhaust-gas treatment unit; said first exhaust-gas treatment unit and said exhaust-gas treatment assembly being arranged axially one after the other in a direction of a longitudinal axis (L.sub.2) of said flow path with said flow path including said first exhaust-gas treatment unit and said exhaust-gas treatment assembly; a hydrocarbon feeding assembly for introducing hydrocarbon into exhaust gas flowing in said flow path; said hydrocarbon feeding assembly including a hydrocarbon-dispensing unit for dispensing hydrocarbon into said flow path downstream of said first exhaust-gas treatment unit and upstream of said exhaust-gas treatment assembly; a swirl-flow-generating unit arranged in said flow path upstream of said hydrocarbon-dispensing unit; a second exhaust-gas treatment unit arranged downstream of said exhaust-gas treatment assembly; and an exhaust-gas/reagent mixing section defining a longitudinal axis (L.sub.3) and having a mixing duct elongated substantially in a direction of said longitudinal axis (L.sub.3) of the mixing section; said exhaust-gas/reagent mixing section further having a reagent-dispensing unit for dispensing reagent into said mixing duct and being disposed downstream of said exhaust-gas treatment assembly and upstream of said second exhaust-gas treatment unit.
